Terraform Task

Task Description:

Launch Linux EC2 instances in two regions using a single Terraform file.

The below Terraform file creates Amazon Linux EC2 instances in Mumbai (ap-south-1) and Singapore (ap-southeast-1) and outputs both public IPs.

```
terraform {
 required_providers {
   aws = {
     source  = "hashicorp/aws"
     version = "~> 5.0"
   }
 }
}

provider "aws" {
 alias  = "mumbai"
 region = "ap-south-1"
}

provider "aws" {
 alias  = "singapore"
 region = "ap-southeast-1"
}

data "aws_ami" "mumbai_server" {
 provider    = aws.mumbai
 most_recent = true
 owners      = ["amazon"]

 filter {
   name   = "name"
   values = ["al2023-ami-*-x86_64"]
 }
}

data "aws_ami" "singapore_server" {
 provider    = aws.singapore
 most_recent = true
 owners      = ["amazon"]

 filter {
```

```
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }
}

resource "aws_instance" "mumbai" {
  provider      = aws.mumbai
  ami           = data.aws_ami.mumbai_server.id
  instance_type = "t3.micro"

  tags = {
    Name = "Mumbai-server"
  }
}

resource "aws_instance" "singapore" {
  provider      = aws.singapore
  ami           = data.aws_ami.singapore_server.id
  instance_type = "t3.micro"

  tags = {
    Name = "Singapore-server"
  }
}

output "public_ips" {
  value = {
    mumbai    = aws_instance.mumbai.public_ip
    singapore = aws_instance.singapore.public_ip
  }
}
```

Steps to get the output:

1. Terraform init

```
sathish@Jeeva-M1 single-file-ec2 % terraform init
Initializing provider plugins found in the configuration...
- Finding hashicorp/aws versions matching "~> 5.0"...
- Installing hashicorp/aws v5.100.0...
- Installed hashicorp/aws v5.100.0 (signed by HashiCorp)

Initializing the backend...


Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

2. Terraform validate

```
sathish@Jeeva-M1 single-file-ec2 % terraform validate
Success! The configuration is valid.
```

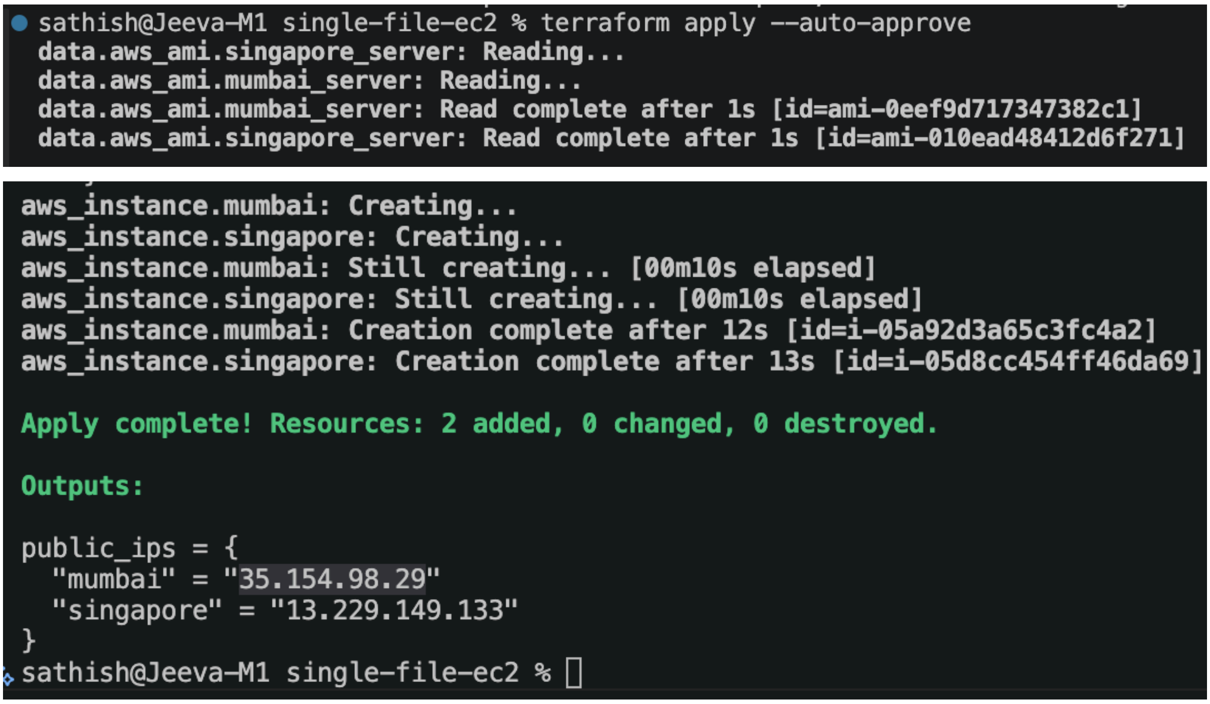
3. Terraform plan

```
sathish@Jeeva-M1 single-file-ec2 % terraform plan
data.aws_ami.singapore_server: Reading...
data.aws_ami.mumbai_server: Reading...
data.aws_ami.mumbai_server: Read complete after 1s [id=ami-0eef9d717347382c1]
data.aws_ami.singapore_server: Read complete after 1s [id=ami-010ead48412d6f271]
```

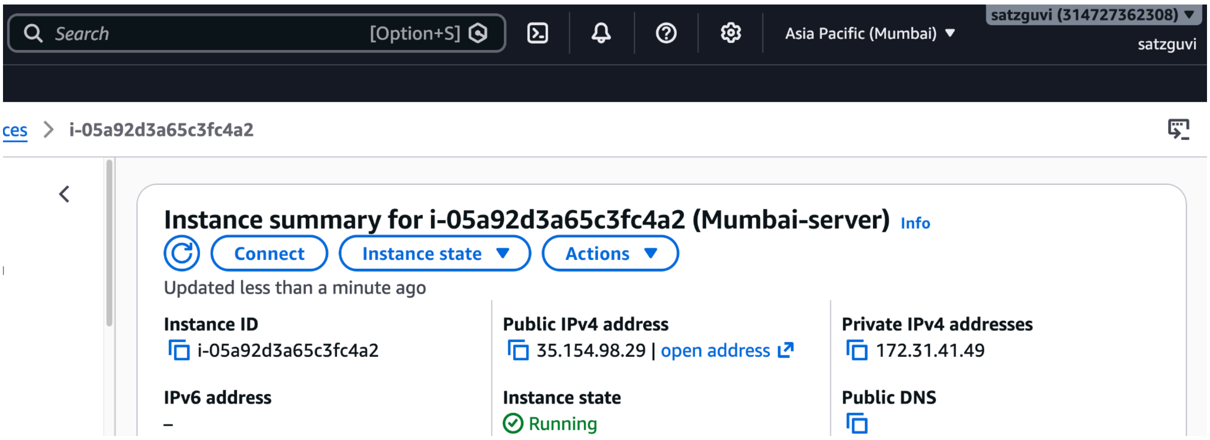
```
Plan: 2 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + public_ips = {
      + mumbai    = (known after apply)
      + singapore = (known after apply)
    }
```

4. Terraform apply



Screenshot of Mumbai instance in AWS console:



Screenshot of the Singapore instance in the AWS console: